

LOGISTIC REGRESSION

$$Pr(y=1 | X, \beta) = f_{\beta}(X)$$

$$Pr(y=0 | X, \beta) = 1 - f_{\beta}(X)$$

$$\Rightarrow Pr(y | X, \beta) = f_{\beta}(X)^y (1 - f_{\beta}(X))^{1-y}$$

Likelihood:

$$Pr(y) = \prod_{i=1}^n Pr(y^{(i)} | X^{(i)}, \beta) = \prod_{i=1}^n f_{\beta}(X^{(i)})^{y^{(i)}} (1 - f_{\beta}(X^{(i)}))^{1-y^{(i)}}$$

Log likelihood:

$$l(\beta) = \log Pr(y)$$

$$= \sum_{i=1}^n y^{(i)} \log(f_{\beta}(X^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\beta}(X^{(i)}))$$

Ta có: $\max_{\beta} l(\beta) \Leftrightarrow \min_{\beta} -l(\beta)$

1, C/M: $l(\beta) = \frac{1}{2} (y - \hat{y})^2$ với $\hat{y} = \frac{1}{1 + e^{-\beta x}}$ không là hàm lồi.

Ta có: $\frac{d l(\beta)}{d \beta} = \frac{d l(\beta)}{d \hat{y}} \frac{d \hat{y}}{d \beta}$

$$= (\hat{y} - y) \times \hat{y} (1 - \hat{y}) = x (-\hat{y}^3 + \hat{y}^2 y + \hat{y}^2 - \hat{y} y)$$

$$\Rightarrow \frac{d^2 l(\beta)}{d \beta^2} = x (-3\hat{y}^2 + 2\hat{y} y + 2\hat{y} - y) \times \hat{y} (1 - \hat{y})$$

$$= x^2 \hat{y} (1 - \hat{y}) (-3\hat{y}^2 + (2y + 1)\hat{y} - y)$$

Khi $\hat{y} \rightarrow \frac{1}{2}$: $-3\hat{y}^2 + (2y + 1)\hat{y} - y \rightarrow -\frac{1}{4} < 0$



HÒA BÌNH

2) (1M): $\mathcal{L}(\beta) = -\sum_{i=1}^n (y^{(i)} \log(\hat{y}^{(i)}) + (1-y^{(i)}) \log(1-\hat{y}^{(i)}))$
 với $y^{(i)} = \frac{1}{1+e^{-\beta^T x^{(i)}}}$ là hàm logit.

Xét $\mathcal{L}(\beta)^{(i)} = -(y^{(i)} \log(\hat{y}^{(i)}) + (1-y^{(i)}) \log(1-\hat{y}^{(i)}))$

$$\frac{d\mathcal{L}(\beta)^{(i)}}{d\beta} = \frac{d\mathcal{L}(\beta)^{(i)}}{d\hat{y}} \cdot \frac{d\hat{y}}{d\beta}$$

$$= \left(\frac{-y^{(i)}}{\hat{y}^{(i)}} + \frac{(1-y^{(i)})(-1)}{1-\hat{y}^{(i)}} \right) \times \hat{y}^{(i)}(1-\hat{y}^{(i)})$$

$$= (\hat{y}^{(i)} - y^{(i)}) x^{(i)}$$

$$\Rightarrow \frac{d^2 \mathcal{L}(\beta)^{(i)}}{d\beta^2} = \hat{y}^{(i)}(1-\hat{y}^{(i)}) x^{(i)} x^{(i)T}$$

Mà $\hat{y}^{(i)}(1-\hat{y}^{(i)}) > 0$ nên $\mathcal{L}(\beta)^{(i)}$ positive definite
 $\Rightarrow \mathcal{L}(\beta)^{(i)}$ là hàm lồi
 xét dương

Mà $\mathcal{L}(\beta)$ là tổng các hàm lồi nên nó cũng lồi.